

07320

DSC02
ALL**Jharkhand University of Technology, Ranchi****Diploma 1st Semester Examination, 2023 (NEP)****Subject : Engg. Chemistry****Subject Code : DSC02****Time Allowed : 3 Hours****Full Marks : 70***Answer in your own words.**Answer five questions in which Question No. 1 is compulsory and answer any four from the rest questions.**All questions carry equal marks.***1. Following questions have four alternative answers. Choose the correct one.****2×7=14**

(i) Two or more atoms having similar mass number are called _____.

(a) isotopes

~~(b) isobars~~

(c) isotones

(d) None of these

(ii) _____ is/are the example of electrolyte.

(a) CuSO_4 (b) NaCl (c) MgCl_2 ~~(d) All of these~~

(iii) _____ is/are used as protective coating, against corrosion.

(a) Galvanization

(b) Tinning

(c) Paint

~~(d) All of these~~

(iv) _____ is not an example of thermoplastic.

(a) Polyethylene

(b) Polyvinyl chloride

~~(c) Polystyrene~~

(d) Bakelite

(v) _____ is a monomer of natural rubber.

~~(a) Isoprene~~

(b) Isotope

(c) Bakelite

(d) Propene

(vi) Which of the following causes air pollution?

(a) Internal combustion engines

(b) Thermal power stations

(c) Burning of fossil fuels

~~(d) All of these~~

(vii) Which of the following is/are greenhouse gases?

(a) CH_4 (b) CO_2 (c) H_2O vapour~~(d) All of these~~

2. (a) Discuss Bohr's atomic theory. Mention its limitations.
(b) What are four quantum numbers? —Explain. 7+7
3. (a) Describe electrochemical series for cations and anions.
(b) What do you understand by corrosion? Discuss different types of corrosion with their mechanism. 7+7
4. (a) Explain different types of polymerisation reactions with examples.
(b) Discuss the process of vulcanisation of rubber with reaction and mention the applications of vulcanized rubber. 7+7
5. (a) Discuss air pollution. What are the sources and effects of air pollution?
(b) What are the causes and control measures of ozone layer depletion? 7+7
6. (a) Discuss electrovalent, covalent and coordinate bond with examples.
(b) What are the purposes of applying paints? Write the characteristics and ingredients of paints. 7+7
7. Write short notes on *any four* of the following: 3.5×4=14
- (a) Arrhenius theory of ionisation
 - (b) Galvanisation
 - (c) Aufbau principle
 - (d) Hydrogen bonding
 - (e) Preventive Environmental Management
 - (f) Deforestation
 - (g) E-Waste
-